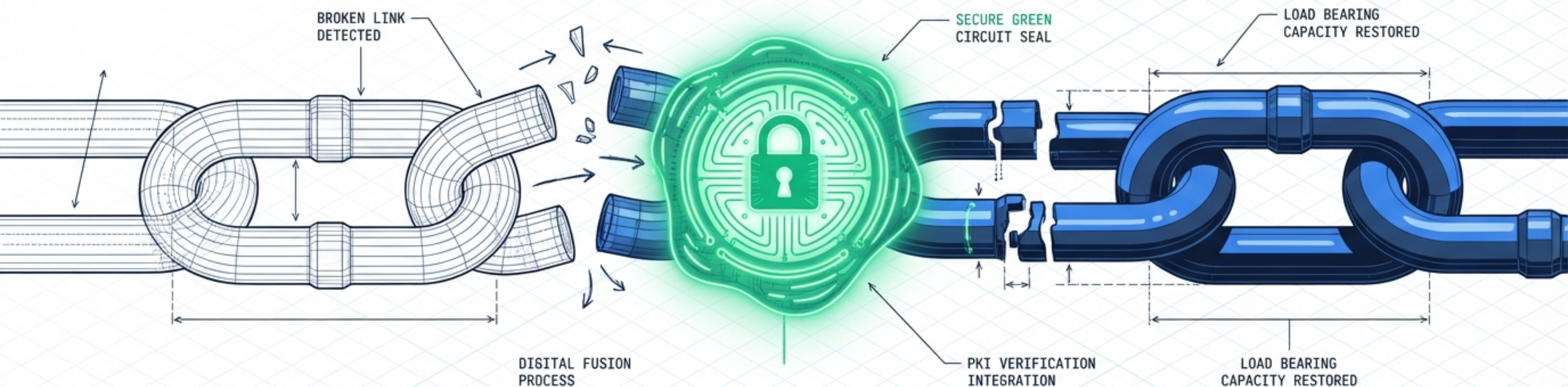


External File Security Services

The PKI-Verified Scanning Chain Architecture (v0.6.14)



Date: 23 Feb 2026
From: Human (Project Lead)
To: Architect (Lead), AppSec, Developer
Status: Architecture Brief – Zoomed-in from File Scanning Pipeline

End-to-End Encryption Creates a Security Blind Spot

Malicious Sender



Untrusted Source

Server



The Blind Courier
Constraint: Server never
sees cleartext.

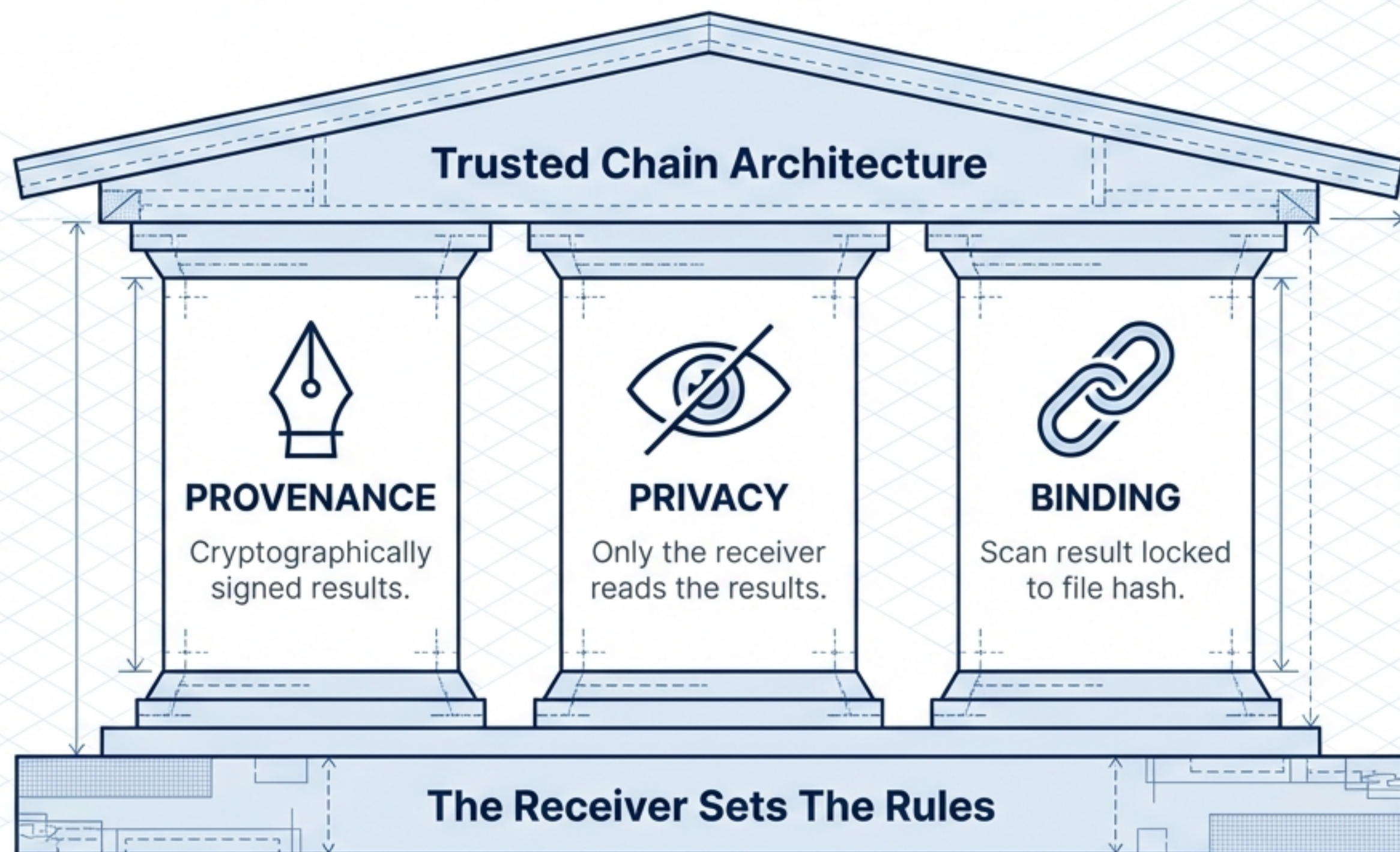
Vulnerable Receiver



The Target
Risk: Client receives
unverified payload.

The Gap: We need a way to trust a scan result
without trusting the person who requested it.

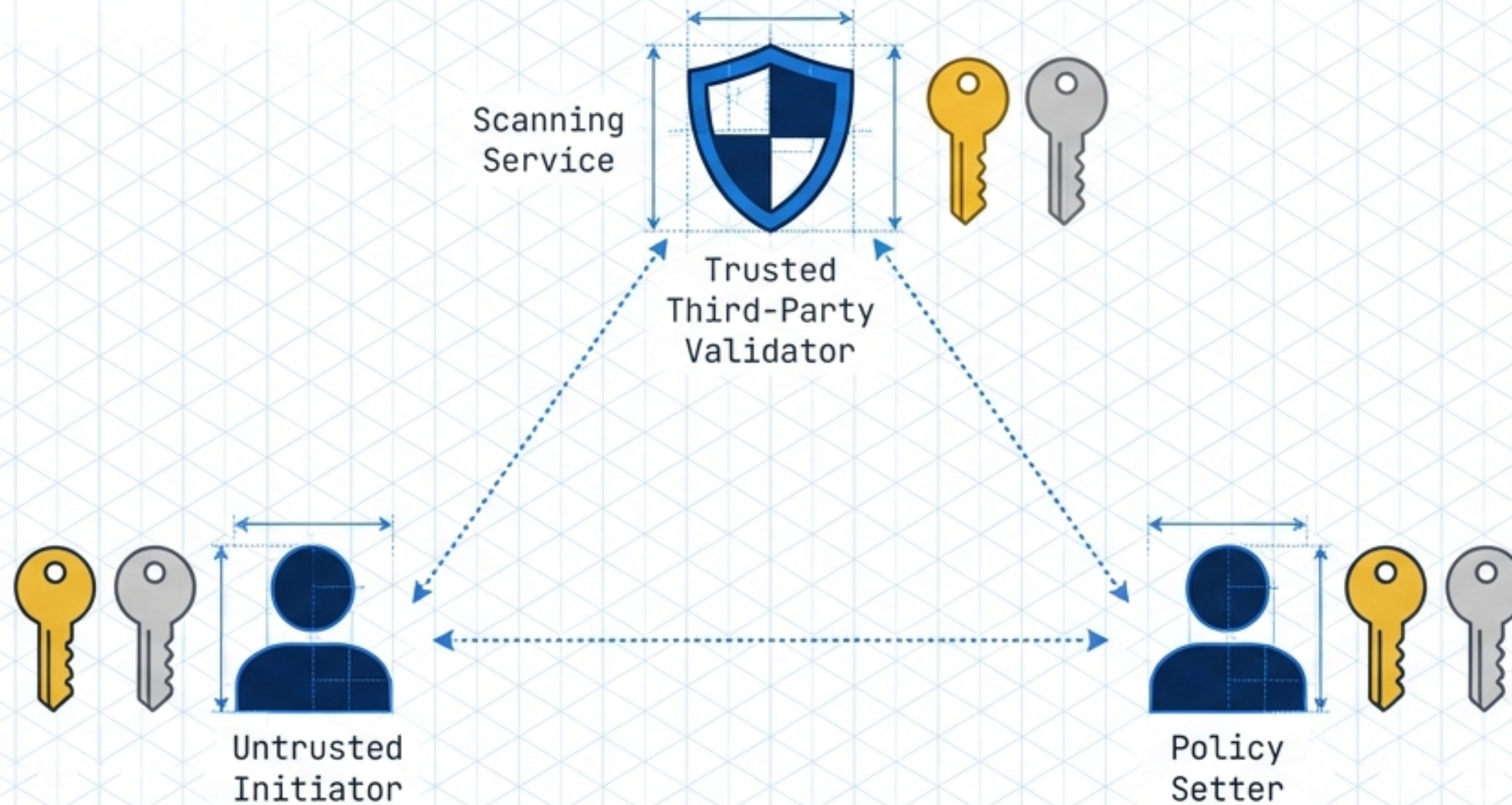
Trust Through Mathematics, Not Reputation



The Core Insight:

The Receiver dictates security.
"Do not send me X unless it is
verified by Y."

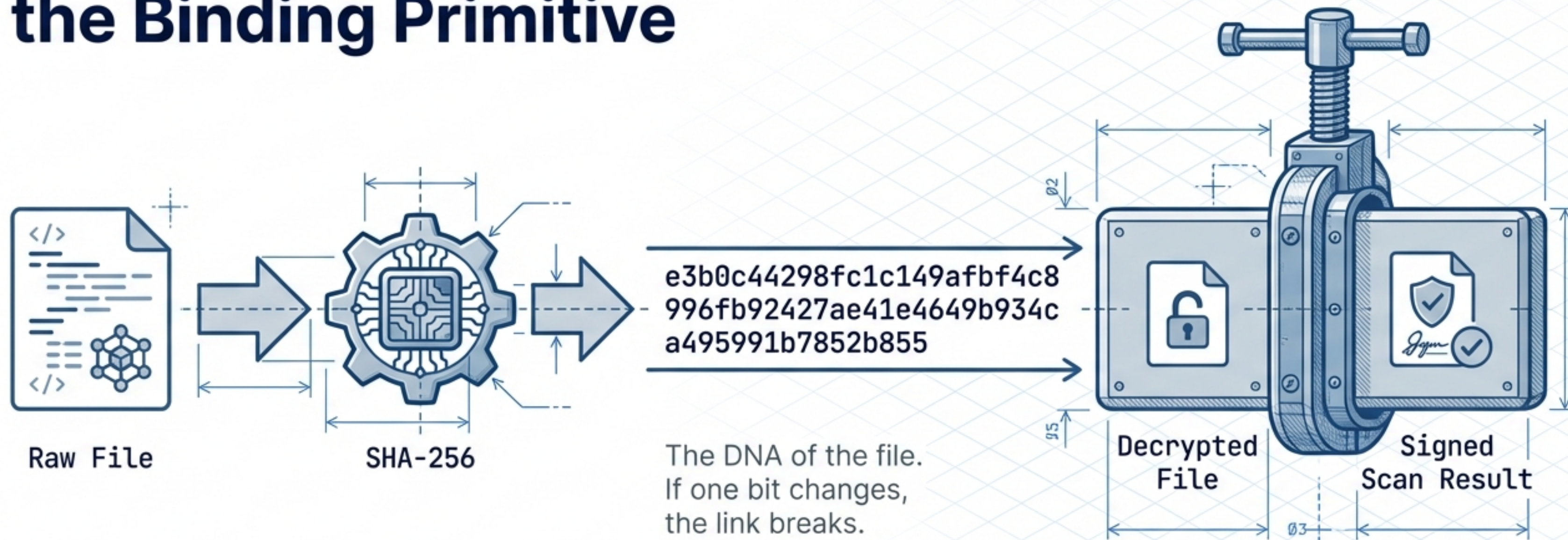
A PKI-Based Ecosystem of Trust



The Rules:

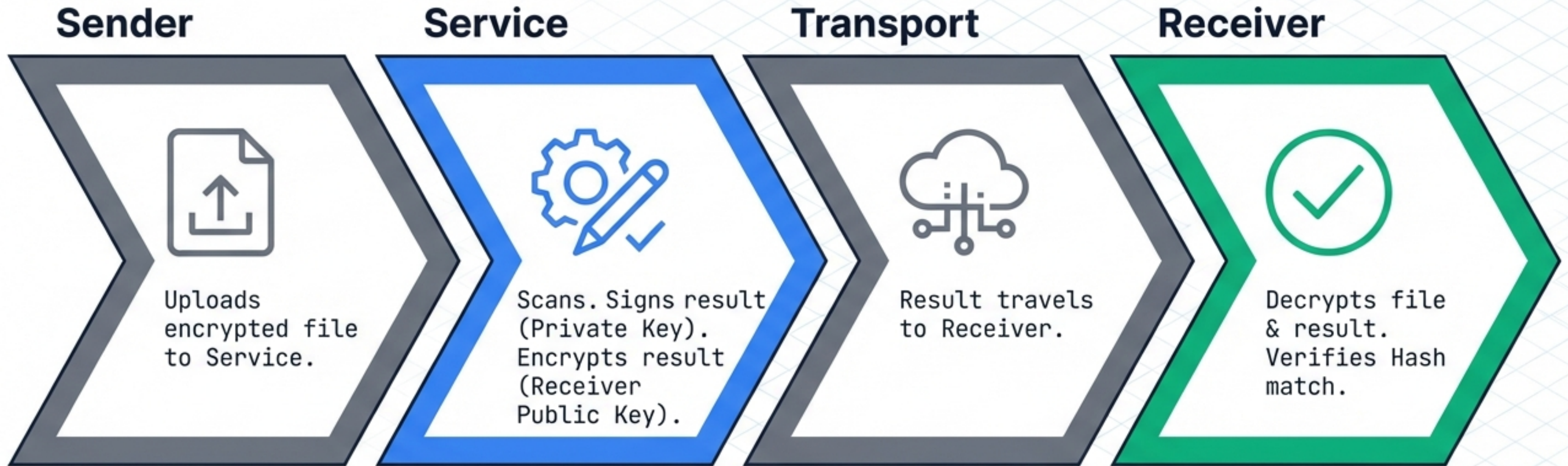
1. Every party has a PKI key pair.
2. Every message is signed (Integrity).
3. Sensitive data is encrypted (Privacy).

The SHA-256 Hash is the Binding Primitive



Privacy Note: Knowing the hash = Knowing the file.
The hash is never shared with the Server.

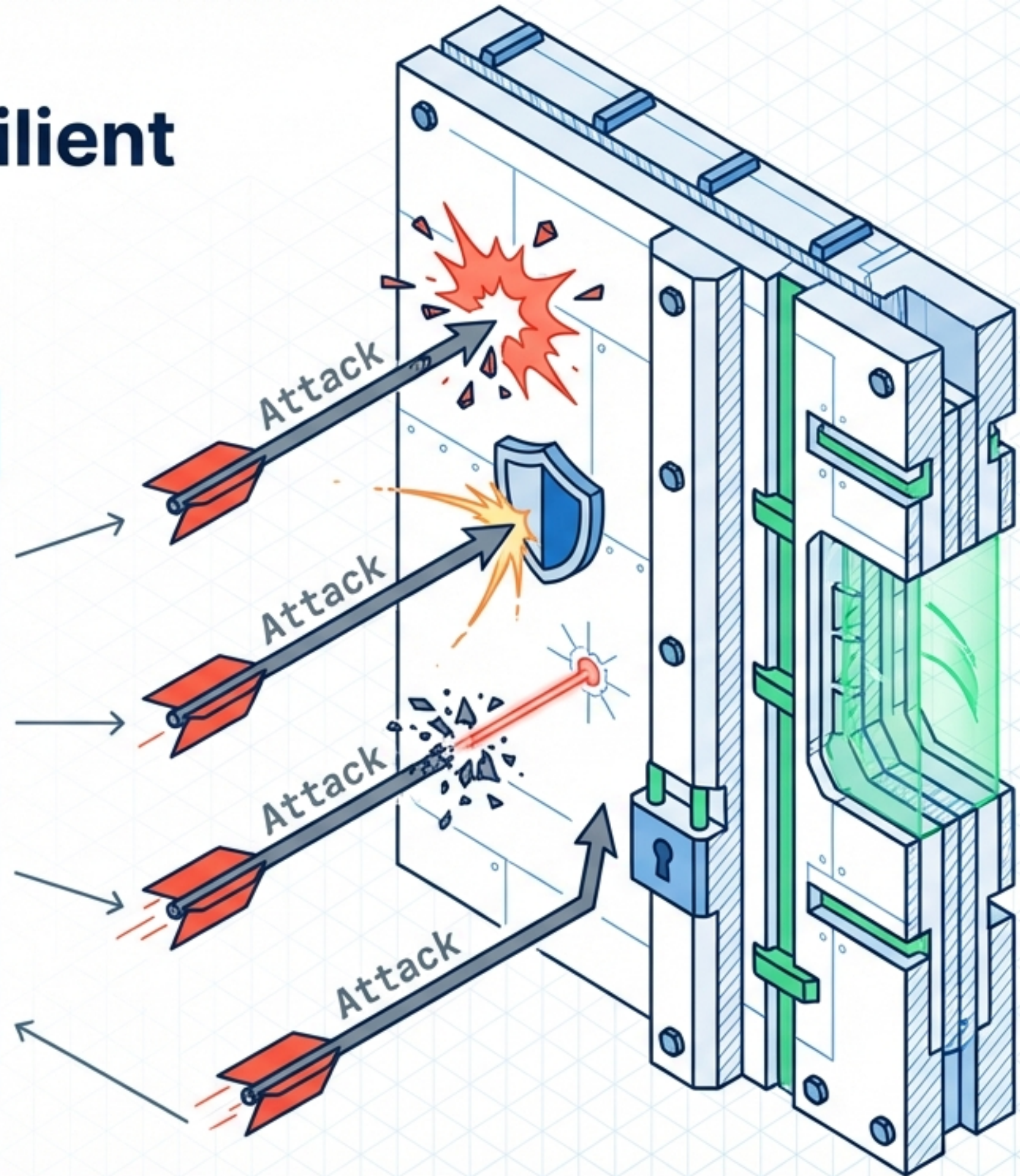
The Linear Journey of Verification



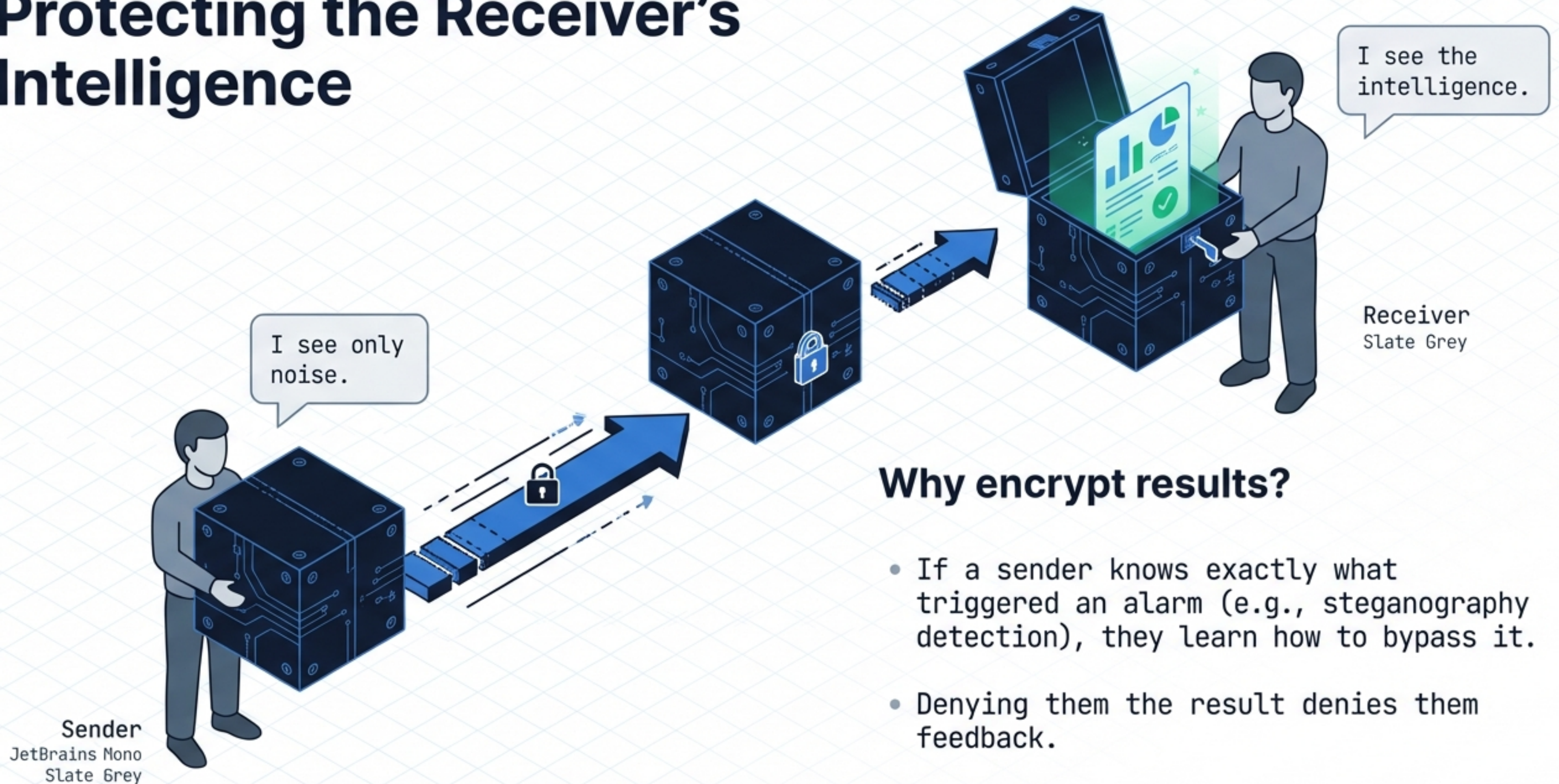
Outcome: If any check (Signature or Hash) fails, the browser blocks the file immediately.

The Architecture is Resilient to Malicious Senders

Attack Vector	Defense Mechanism
Skip scanning?	Receiver demands result; browser rejects file.
Forge the result?	Sender lacks Service's Private Key.
Swap file after scan?	Hash mismatch at Receiver end.
Read the result?	Encrypted for Receiver only.



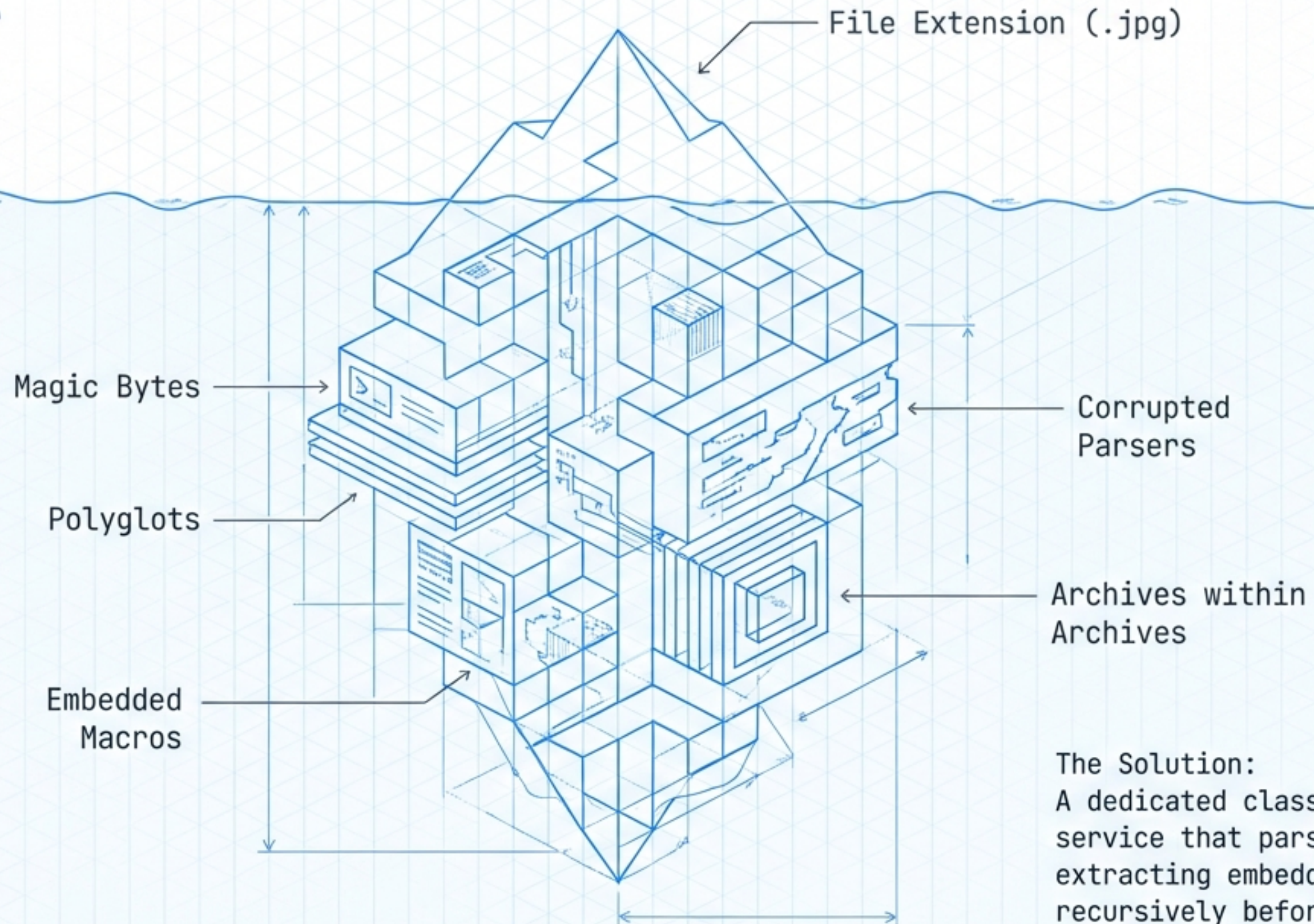
Protecting the Receiver's Intelligence



Why encrypt results?

- If a sender knows exactly what triggered an alarm (e.g., steganography detection), they learn how to bypass it.
- Denying them the result denies them feedback.

'File Type' is an Opinion, Not a Fact



The Solution:
A dedicated classification
service that parses structurally,
extracting embedded content
recursively before declaring a type.

Chaining Services and the 'Pod' Model

The Chain



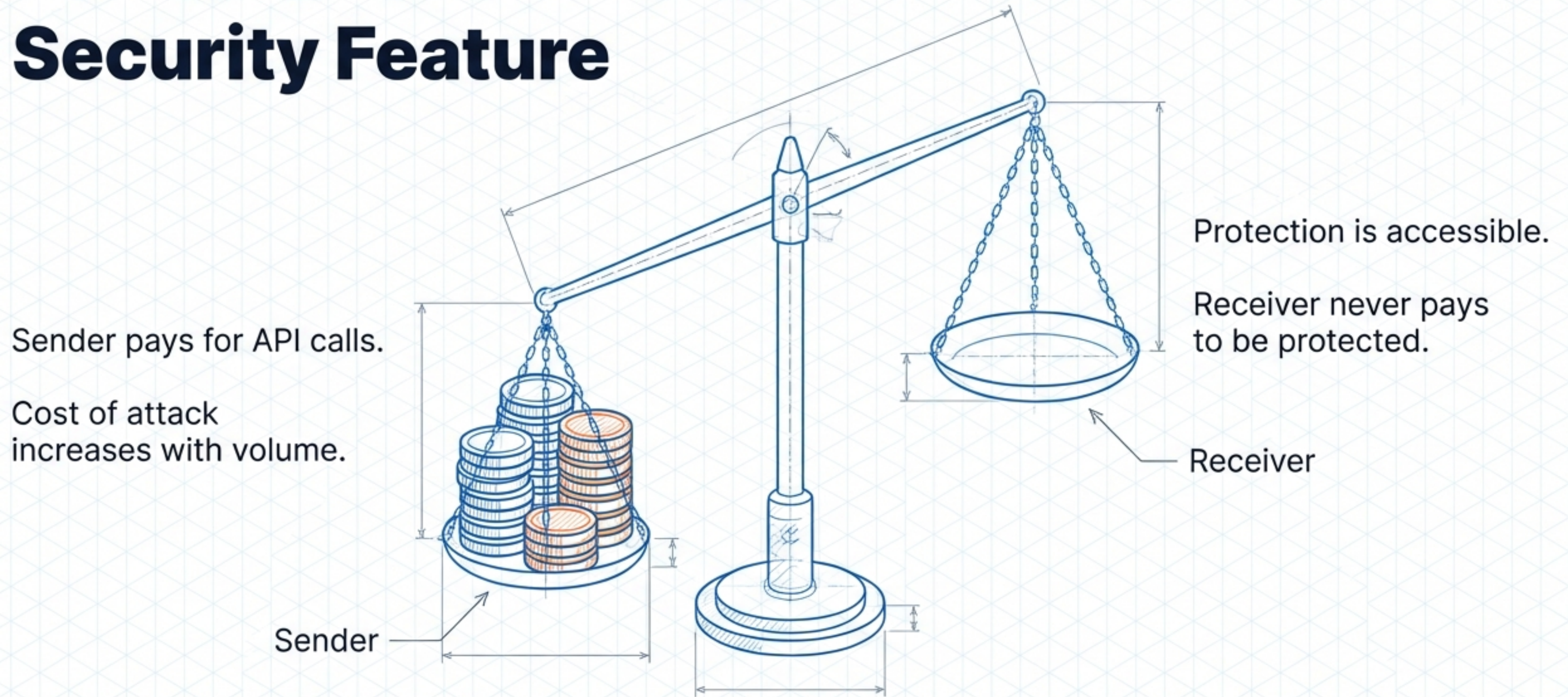
The Pod



High Assurance Variant: The Pod Intermediary

- Benefit: Receiver knows a trusted intermediary processed the file.

Economic Friction as a Security Feature



Impact: Anti-spam by design.

Non-Repudiation and Audit Trails

Swiss International Engineering



Complete Auditability:

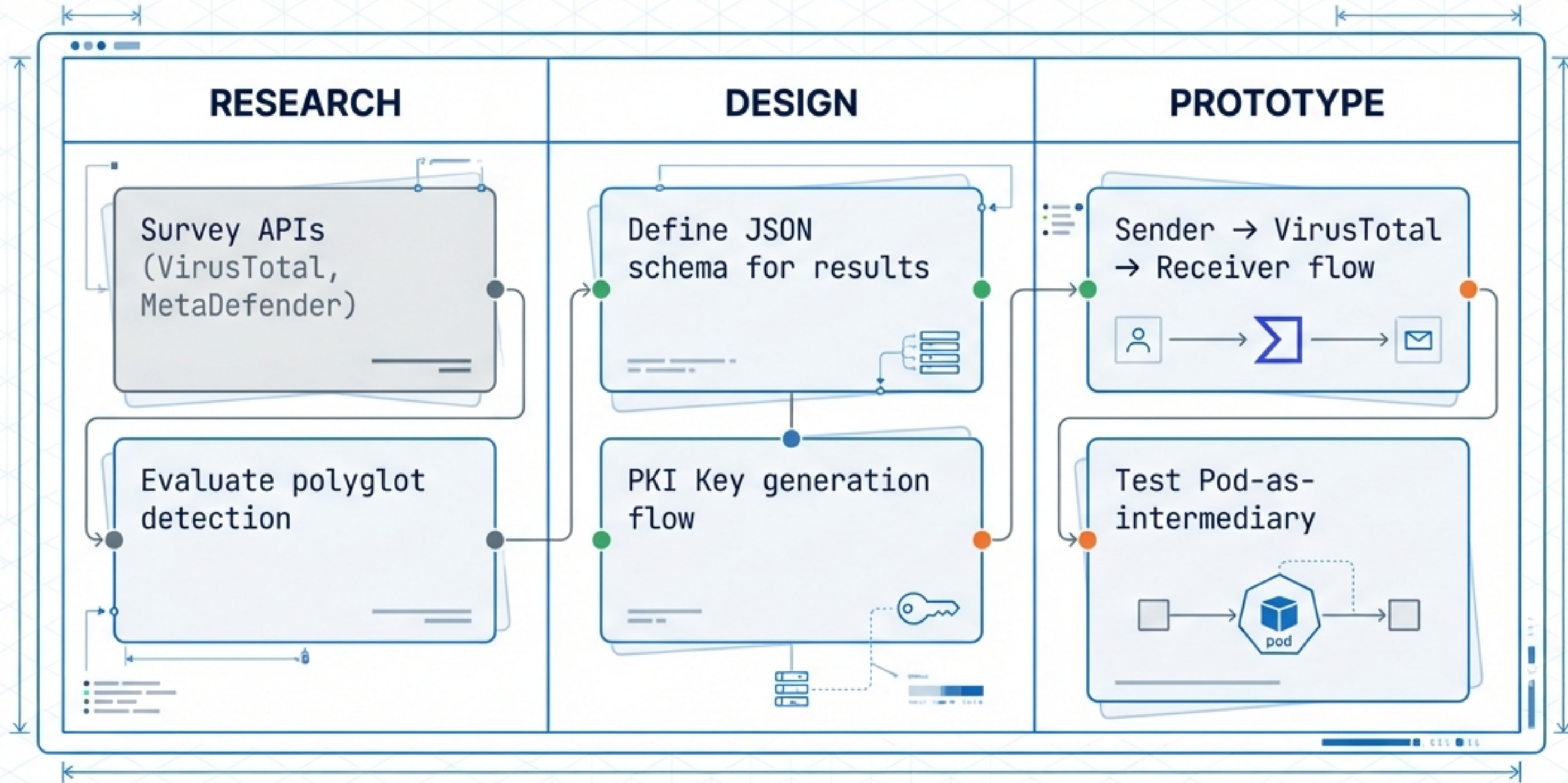
- We know Who scanned it, When, and What the result was.
- No party can deny their actions later.

Discovery and the Web of Trust

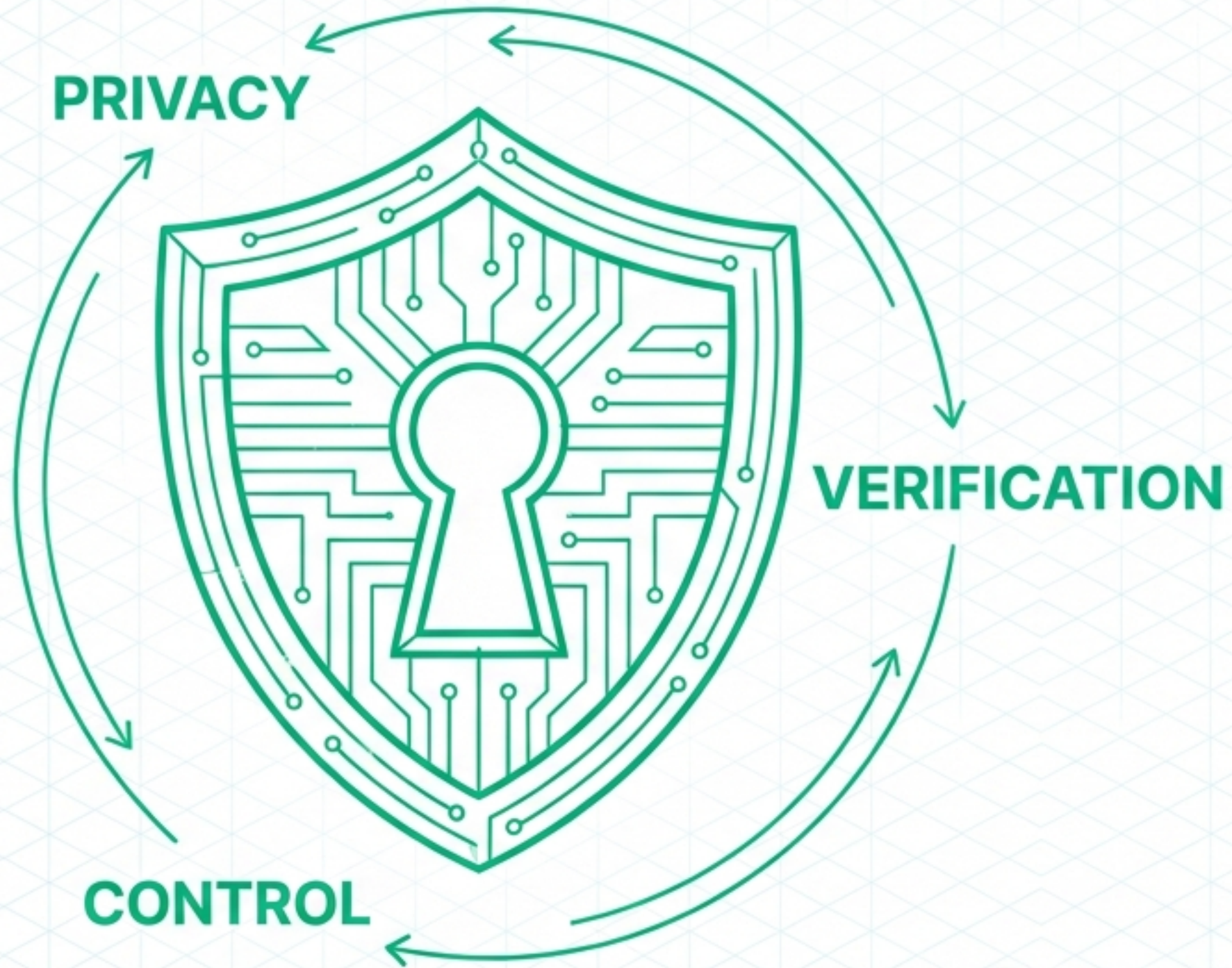


- **Setting 1 (MVP):**
Platform maintains list of approved keys.
- **Setting 2 (Future):**
Receiver manually adds trusted keys.
- **Setting 3 (Advanced):**
Algorithmic trust based on accuracy.

Roadmap: From Theory to Prototype



Enabling Trust in Zero-Trust Environments



- **Problem Solved:**
The 'Blind Server' gap is closed.
- **Privacy Kept:**
E2EE remains intact.
- **Power Shifted:**
The Receiver holds the keys to safety.

We verify, so we do not have to trust.